Branch Coverage Report

Decision (branch) coverage of the vps package | 6 November 2026

1. Summary

Branch coverage 93.7% (251 of 268 branch outcomes). Every if, loop and conditional expression has two outcomes; a branch counts as covered only when a test took that outcome. Environment: Windows 11, Python 3.14, pytest 9 with pytest-cov, run on 6 November 2026 against the main branch. The same suite runs on every pull request in GitHub Actions on Ubuntu, Windows and macOS with Python 3.10 and 3.12.

2. Branch coverage by module

Module	Branches	Covered	Partial/missed	Branch coverage
__init__.py	6	5	1	83.3%
admin.py	54	47	7	87.0%
audit.py	0	0	0	no branches
auth.py	32	31	1	96.9%
clock.py	0	0	0	no branches
db.py	6	6	0	100.0%
demo.py	22	19	3	86.4%
fees.py	6	6	0	100.0%
grid.py	42	40	2	95.2%
parking.py	30	29	1	96.7%
payment.py	6	5	1	83.3%
prediction.py	8	8	0	100.0%
reports.py	26	25	1	96.2%
seed.py	14	14	0	100.0%
slots.py	16	16	0	100.0%
Total	268	251	17	93.7%

3. Key decisions and the tests that take both outcomes

Decision	Covered by
Stay within grace period or not (fees.billable_hours)	test_exit_fee: 4:59, 5:00, 5:01
Preferred slot still free and reachable (grid.assign_slot)	test_route: kept, taken, blocked
Plate valid or not (parking.normalise_plate)	test_entry: valid, short, long, symbols
Forced payment failure on or off (payment.process_payment)	test_payment: both modes
Account locked, wrong or right password (auth.attempt_login)	test_security: 4, 5 failures, unlock
Role allowed or denied (auth.role_required)	test_auth, test_admin: admin and attendant
Enough history for an estimate (prediction.estimate_minutes)	test_prediction: 4 and 5 records
CSRF token missing, forged or valid	test_security: all three

4. How to reproduce

python -m pytest --cov=vps --cov-branch --cov-report=term-missing